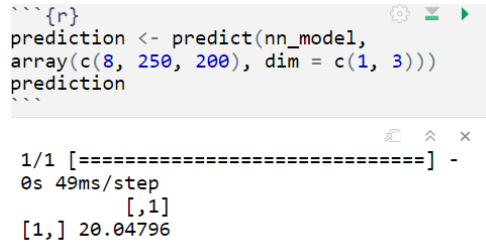


Module 02: Lab 02

Exercises

1) Using the neural network above, what is the predicted miles per gallon of a vehicle with 8 cylinders, a displacement of 250, and a horsepower of 200?

```
## {r}
prediction <- predict(nn_model,
  array(c(8, 250, 200), dim = c(1, 3)))
prediction
```



```
1/1 [=====] -
0s 49ms/step
      [,1]
[1,] 20.04796
```

The predicted miles per gallon was 20.058 miles per gallon.

2) What loss function did we specify when building the neural network model above?

We specified the loss function to be the mean squared error. This ensures that our optimization of our neural network and penalized large errors more heavily. This helps the training of the neural network.

3) Define the rank of a tensor. What is the rank of a matrix (remember, a matrix is a tensor of a particular rank)?

The rank of a tensor is defined by the numbers of dimensions. The rank of a matrix is two (rows and columns) it has 2 dimensions.

4) The data we will be working with throughout the course, including in this lab, is vector data. What is the tensor rank of vector data?

The tensor rank of a vector data is 1.

5) Consider the following tensor of ones: array(1, dim = c(500, 256, 256, 3)). What is its rank? What is its shape? What is the dimension of the second axis? If this tensor represented a collection of color images, how many images are there?

The rank of our tensor is 4, it has 4 dimensions.

The shape of our tensor is 500, 256, 256 and 3 respectively.

The dimension of the second axis is 256 respectively.

If our tensor represented a collection of color images it is represented by the first dimension which is 500.